Appendix: Four Views of the Strangest Experiment

In 1978, not long after moving to the University of Texas, John Wheeler proposed a thought experiment that, according to him, "gets at the core of what fueled the Bohr-Einstein debate." In fact, he suggested that "this experiment may have something to tell us about the very machinery of the universe." He called it the delayed-choice experiment (Figure A.1).

The experiment has two configurations; we'll start with the simpler one, on the left (Figure A.1a). A laser beam (i.e., a beam of photons) enters from the lower left-hand corner and enters a beam splitter, which (as the name suggests) splits the beam into two equal parts: one gets bounced up and one passes straight through to the right. The two beams each hit one more mirror, setting them on a course to cross paths again. Each beam hits a photon detector, and that's it.

Now, consider the same experiment, but with a twist (Figure A.1b, on the right). At the point where the two beams cross in the upper right-hand corner, before they hit the detectors, place a second beam splitter. Each of our two beams splits again: now half of each beam goes off to the right, toward detector 2, and half of each beam goes up, to detector 1. But the beam splitter is constructed in a funny way, so the two combined half-beams don't behave the same way in both directions. The two half-beams going up are in sync: their peaks and valleys line up with each other, reinforcing their combined wave. This is constructive interference, like the bright bands in the double-slit experiment in Chapter 5. But the two half-beams going off to the right are perfectly out of sync: the peaks of one line up with the valleys of the other, so they totally cancel out. This is destructive interference, like the dark bands in the double-slit experiment.